CST523 - Compiler Construction

CST 523 (Compiler Construction) Syllabus

Course Information

Course Details

- **Course Title:** Compiler Construction
- **Course Number:** CST 523
- **Meeting Location:** Physical
- **Units:** 2

Course Description

This course covers the principles of compiler design, focusing on front-end components such as lexical analysis, syntax analysis, parsing, semantic analysis, and code optimization. Emphasis is placed on translating high-level programming languages into intermediate representations. Students will gain practical experience in constructing front-end components of simple compilers through hands-on projects.

Materials

- Course textbook: "Compilers: Principles, Techniques, and Tools" by Alfred V. Aho, Monica S. Lam, Ravi Sethi, and Jeffrey D. Ullman

Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system

Course Objectives

At the end of this class, you should be able to:

1. Design and implement lexical analyzers and parsers for programming languages
2. Perform syntax and semantic analysis on source code
3. Translate high-level language constructs into intermediate representations
4. Apply code optimization techniques in compiler front-end design

Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

Assignment Kind	Value
Projects	35%
Exams	35%
Participation	30%

Assignment Kind	Value

Projects

Projects will take the form of implementing a simple lexical analyzer, building a basic parser for a small language subset, and creating a semantic analyzer for basic language constructs. Students will focus on understanding compiler front-end fundamentals through hands-on implementation of core components.

Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.